

Mubarek Yeshaw Tahir

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Skills

Programming Languages: C++, Python, Java, Rust

Technologies: MATLAB, Microsoft Azure, Networking (TCP/IP, DNS, DHCP)

Certifications: Cisco CCNA | *Pursuing: Microsoft Certified: Azure Administrator Associate*

Languages: Amharic (native), English (C1), German (B1)

Experience

Junior Network Administrator (IT Intern), Alkan CIT – Addis Ababa, Ethiopia Summer 2022, Summer 2023

- Managed and optimized company network infrastructure, configuring routers and switches and resolving TCP/IP, DNS, and DHCP issues
- Implemented security protocols and monitored network traffic; deployed and administered Microsoft Teams infrastructure for enterprise clients
- Trained staff on network best practices to improve team efficiency

Projects

Safe Rust Implementation of the RaSTA Railway Protocol – Advanced Security Engineering Lab, University of Passau

- Contributed to standards analysis (DIN VDE V 0831-200, EN 50159 fault model) and to the `#![no_std]` Rust implementation (core protocol state machine and platform adapters) of the RaSTA SIL 4 railway safety protocol, as part of a 4-person team
- Achieved 92.78% line coverage across 181 automated tests with zero unsafe blocks, passing all CI quality gates including Clippy with `-D warnings`
- Ran interoperability testing against the SBB TÜV-validated reference stack across 6 test scenarios, measuring Rust-to-SBB round-trip times of 18.49 μ s median (n = 5,000)

Secure Watch: Violence Detection and Alert System (VDAS) – Bachelor's Thesis, Adama Science and Technology University

- Co-developed an AI-powered violence detection system for CCTV surveillance as part of a 5-person team, using a pretrained EfficientX3D model for real-time action classification
- Contributed to dataset collection (Kaggle Real-Time Violence Detection dataset, 2,000 labeled videos) and model building/training
- Delivered as a desktop application and Progressive Web App with real-time CCTV monitoring and automated security-officer alerting

Azure Cloud Infrastructure – Personal Project

- Set up a multi-VNet Azure environment with network security groups (NSGs) to enforce Zero-Trust-style access control between subnets, with monitoring configured via Azure Monitor

Education

University of Passau – Master of Science in Informatics

Expected March 2028

- Relevant coursework: Wireless Security, Hardware-oriented Security, Energy Informatics

Adama Science and Technology University – Bachelor of Science in Computer Science and Engineering

July 2024

- Relevant coursework: Computer Networks, Wireless Mobile Networks, Distributed Systems, Computer Systems Security